Nucleus Instance Segmentation and Cancer Identification using Machine Learning

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Abstract

Background: Machine learning algorithms are increasingly being used in the medical field to detect cancer from histopathological images, as manual detection of cancer by expert pathologists and clinicians, is a time-consuming affair. The prognosis is worse when cancer has spread to the lymph nodes. Fortunately, with the rise of digital pathology, high-resolution digital images of histopathological scans of lymph node sections are increasingly available. Manual Assessments of Haemotoxylin and Eosin (H&E) suffer from indistinct guidable observability. To address this problem whole slide images (WSI) are acquired from glass Histopathology slides using a scanning device that analyzes the tissue specimen. The extraction of the nucleus will enhance the diagnostic potential of computation pathology as the WSI contains a significant number of clustered nuclei in which tumors are present, i.e., correctly segmenting the nucleus present in the cluster will allow for downstream analysis of tumor infiltration lymphocytes (TILS).

Methods: To achieve roto-translation invariance in the features extracted from histopathological images, we employed group equivariant convolutions. This approach allows the full utilization of the symmetrical properties inherent

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in histopathological images through the implementation of a group equivariant convolution network(G-CNN). the classification model is built using G-CNN

Results: The utilization of deep learning models incorporating G-CNN layers has significantly enhanced the ability to detect metastatic features within the Pcam dataset. This is evident from the final AUC score of 0.98. In addition, for the task of identifying nucleus instances, an encoder-decoder architecture employing G-CNN layers has been employed, resulting in a Dice value of 0.64 and a PQ value of 0.25.

Conclusion: With the comparison of group equivariant convolution network and convolution network concluded that the use of GNN-based models for histopathology gives better results in both classifications as well as nucleus instance segmentation.

Keywords: Deep learning, Group Equivariant Convolution, Nucleus Instance Segmentation

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1. Introduction

Various forms of convolution neural networks with non-identical approaches to learning the parameters of images have attracted a lot of attention. Group Equivariant Convolutional Networks (G-CNNs) refer to a type of convolutional neural networks where convolution can be defined for any group G. This group G can be represented through group rotations, reflections, and translations of the pixel grid. G-CNNs can be seen as a generalization of traditional CNNs, providing a significant increase in the network's expressive capacity without requiring an increase in the number of parameters [1]. Equivariance and invariance are crucial properties for various computer vision problems.

Images have complex features to be learned and require deep neural networks to perform well on vision problems. Various researchers have leveraged the pre-trained DL model[2] to classify breast cancer related to the BreaKHis histopathology images For instance, VGG16 and AlexNet[3] as pre-trained DL models are used by authors in [4] to classify the Breakhis dataset. Achieving

rotation-invariance is a crucial property for machine learning models in medical image analysis, particularly in computational pathology applications. To address this, a framework has been proposed that encodes the geometric structure of the Special Euclidean motion group SE(2) [5] in convolutional networks, resulting in translation and rotation equivariance through the use of SE(2)-group convolution layers. This allows models to learn feature representations that have a discretized orientation dimension, ensuring that their outputs are invariant under a discrete set of rotations. While conventional approaches to rotation invariance rely mainly on data augmentation, this does not guarantee the robustness of the output when the input is rotated [6]. Thus, conventional CNNs may require test-time rotation augmentation to reach their full potential [7]. This study specifically focuses on histopathology image analysis applications where machine learning models mustn't capture arbitrary global orientation information of the imaged tissues [5]. By allowing convolutional neural networks (CNNs) to be rotation-equivariant, the need to learn the set of transformations from data is eliminated, allowing for more discriminative features to be learned and freeing up model capacity [8]. Additionally, reducing the number of required parameters decreases the risk of overfitting. Dense Steerable Filter CNNs (DSF-CNNs) [9], which employ group convolutions with multiple rotating copies of each filter in a densely connected framework. In contrast to traditional filters, each filter is defined as a linear combination of steerable basis filters, giving precise rotation and reducing the number of trainable parameters. the two independent computational pathology tasks which are breast tumor classification and nucleus instance segmentation solved efficiently if the model architecture incorporates group equivariant convolution.

2. Methodology

This section explains the problem formulation, the complete description of group equivariant convolution, with group formation, and the methods used to formulate the binary classification model and the nucleus instance segmentation

model.

2.1. Dataset

The two publicly available datasets chosen, for the proposed work. The datasets used are as follows:

- *PatchCamelyon (PCam) Dataset*: The *PCam dataset* [10, 11] contains 3,27,680 color images (96 X 96 pixels each) that were taken from lymph node section histopathologic scans. Each image has a binary label that indicates whether metastatic tissue is present. A new benchmark for machine learning models is provided by PCam, which is trainable on a single GPU, larger than CIFAR10, and smaller than imagenet. The dataset is divided into a training set of 2,62,144 (2^18) examples, and a validation and test set both of 32,768 (2^15) examples. There is no overlap in images between the splits, and all splits have a 50:50 balance between positive and negative examples.

- *Kumar Dataset*: The *Kumar Dataset* [12], has been generated by downloading 40x-magnified H&E stained tissue pictures from the TCGA library [13]. For tumor assessment (grading, staging, etc.), H&E staining is a standard procedure to increase the contrast of a tissue section. The training dataset contains generalizable nuclei segmentation approaches that the diversity of nuclei appearances across several organs and patients as well as the richness of staining protocols implemented at numerous hospitals.

2.2. Convolution on group

Convolution operation for groups can be analyzed by the symmetric groups that are used in G-CNNs. A symmetric group after transformation remains invariant. Group formation is completely dependent on groups and consists of a symmetrical transformation of sampling lattice which is translation, rotation, and reflection. this geometrical transformation changes the sampling lattices. for example if we take a image grid Z^2 and flip the grid image can be represented as $-Z^2 = (-x, -y)|(x, y)\ \varepsilon\ Z^2 = Z^2$. this operation is a symmetric

transformation for the group. the equivariance property of convolution can be explained by group structure.

The Group $P4$ The group taken is composed of rotation by 90 degrees about the center of the square grid[8]. its parameters are r, x, y here r = 0, 1, 2, 3 and $(r, x, y)\ \varepsilon\ Z^2$. the group P4 can be represented as

$$g(r,x,y) = \begin{bmatrix} \cos(r\pi/2) & -\sin(r\pi/2) & x \\ \sin(r\pi/2) & \cos(r\pi/2) & y \\ 0 & 0 & 1 \end{bmatrix}$$

The operations are represented by matrix multiplication of the group. this can also be represented directly in terms of equations in tuples which will be complicated. the simplified method is to compose it into two group elements represented by integer tuples converted into the matrix form, by multiplying these matrices. In the context of group $P4$ operates on points in Z^2 (pixel coordinates) by multiplying the matrix $g(r, x, y)$ by the homogeneous coordinate vector $u(x', y')$ of a point (x, y).

$$gx \simeq \begin{bmatrix} \cos(r\pi/2) & -\sin(r\pi/2) & x \\ \sin(r\pi/2) & \cos(r\pi/2) & y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

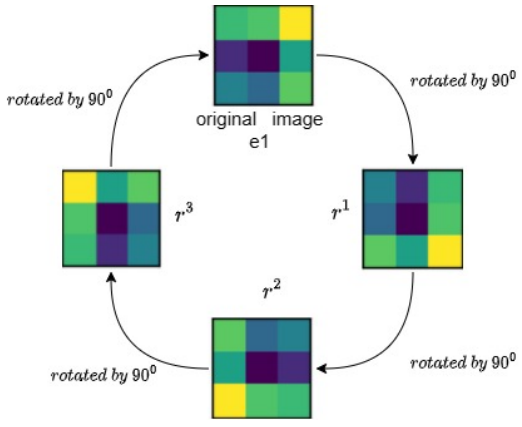


Figure 1: Graphical embodiment for the group p4

The Group $P4m$ In this final group, all transformations are covered like rotation by 90 degrees, reflection, and translation around the square grid. here one extra parameter m is for reflection where $m\ \varepsilon\ 0, 1$.

$$g(m, r, x, y) = \begin{bmatrix} (-1)^m \cos\left(\frac{r\pi}{2}\right) & -(-1)^m \sin\left(\frac{r\pi}{2}\right) & x \\ \sin\left(\frac{r\pi}{2}\right) & \cos\left(\frac{r\pi}{2}\right) & y \\ 0 & 0 & 1 \end{bmatrix}$$

In Convolutional Neural Network (CNN), images and stacks of feature maps

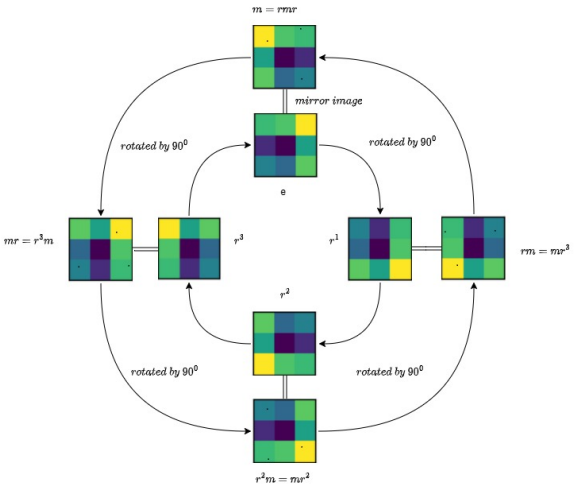


Figure 2: Graphical embodiment for the group p4m

are expressed as functions $f : \mathbb{Z}^2 \to \mathbb{R}^K$, where K represents the number of channels. These functions are defined on a limited domain, typically a rectangular area. For each pixel coordinate (x, y) in the 2D integer lattice $\mathbb{Z}^2$, the feature map stack yields a K-dimensional vector, denoted as $f(x, y)$. the focus lies on transformations applied to these feature maps (taken as a group). This can be formalized by $[Lgf](x) = [f \circ g^{-1}](x) = f(g^{-1}x)$, where g denotes a transformation acting on the set of feature maps. This notation describes the transformed feature maps, with Lg signifying the impact of the transformation g on the function f.

3. Group Equivariant Convolution Layer

The translational equivariance of the convolutional layer implies that the network's output remains unaltered when the object undergoes shifts along the sampling lattice, barring potential edge effects. This directly enables the efficient detection of objects of the same shape.

The convolution layer is rotationally variant if the shifted image by 90 degree is passed through a deep network then the resulting feature maps obtained are different from the feature maps obtained by passing the original input image through the same deep network.

Group Equivariant convolution is a new type of layer that employs a substantially higher degree of weight sharing than regular convolution layers. equivariant convolutions increase the expressive capacity of the network without increasing the number of parameters. the group equivariant convolution is implemented with the help of a combination of two layers. [14].

1. **P4ConvZ2 convolution layer:** This layer introduces an additional dimension to the input, encompassing all the group elements. The resulting structured output feature map, for example, with an input image size of $[32 \times 32]$, becomes $[32 \times 32 \times 4]$. This extension accommodates all 90-degree rotations along with their reflections, adding a crucial dimension to the output.

   To optimize the time complexity involved in calculating rotated ver-

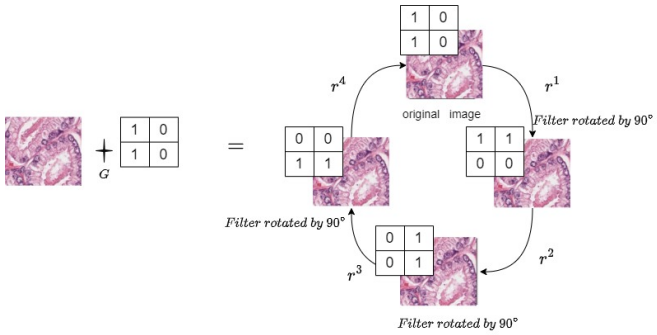


Figure 3: Convolution operation of P4ConvZ2 layer.

sions of the image, a strategy is implemented where the filter undergoes rotation and is convoluted with the original image. This process yields a stack of four output images, effectively capturing the diverse rotated representations.

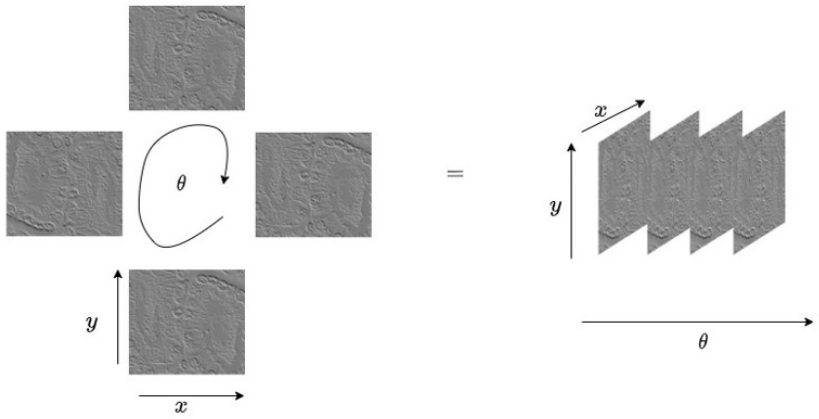


Figure 4: Output given by P4ConvZ2 layer.

2. **P4ConvP4 convolution layer:**

This layer processes a structured feature map as its input, with convolution involving two crucial elements: transformation and the dot product of groups. The transformation comprises two parallel actions: individual rotation of filters and movement from one group element to another. As illustrated in the figure below, the transformation includes four distinct filters that undergo individual rotations and are shifted to other filter positions. Meanwhile, dot product of identical structured objects involves taking the dot product for all the structured elements within the group and aligning them with the respective initial image of the group. This process yields the complete structure of the image, encompassing the four feature maps produced by this layer.

For each group element, a similar operation is performed with its eight

rotated versions of the filters

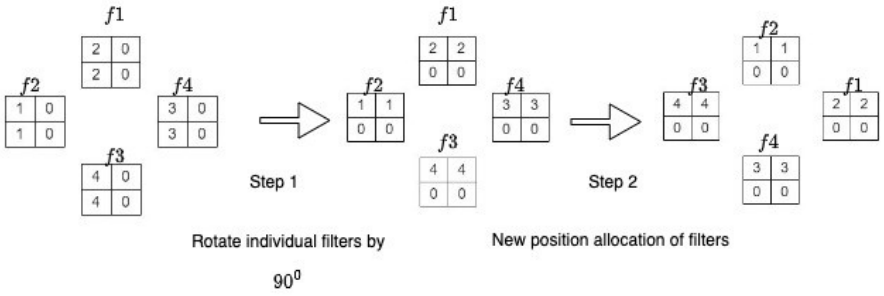


Figure 5: Transformation of filters.

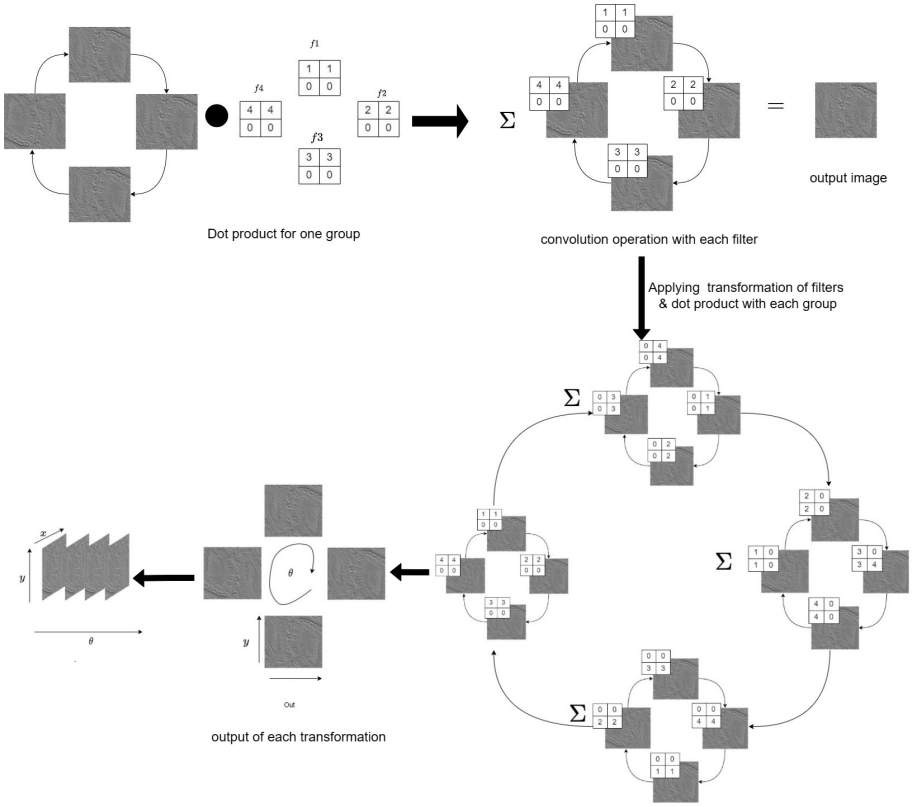


Figure 6: Complete Operation and output of P4ConvP4 layer.

3.1. Model for Classification

The architecture of the G-CNN-based model is structured with the initial layer being the P4mConvZ2 convolutional layer, followed by batch normalization and Rectified Linear Unit (ReLU) activation. This is succeeded by a corresponding structured input to P4mConvP4m, which also incorporates batch normalization and ReLU activation. Additionally, linear layers are incorporated into the design to learn the features. In each fully connected layer (excluding the output layer), Batch normalization and ReLU (in that order) are applied to produce the output, with an additional dropout regularization (with a probability of 0.5) to address overfitting concerns.

The layer names, P4mConvZ2 and P4mConvP4m, signify operations that respect the symmetries of the dihedral group P4m. The "P4m" designation encompasses 90-degree rotations (P4) and mirror reflections (m), and these convolutional layers are designed to be equivariant under these transformations. Specifically, P4mConvZ2 suggests equivariance under translations in two dimensions (Z2), while P4mConvP4m implies equivariance under the transformations of the dihedral group P4m.

In the final model iteration, a learning rate of 0.00005 is utilized, with a batch size of 16 during training. The Adam optimizer is employed for the training process. This architectural framework outlines the arrangement of layers for feature extraction from the image, with the convolutional layers explicitly designed to capture symmetries present in the data.

Parameters: To calculate the parameters in the group equivariant convolution layers, a method similar to the standard convolution approach can be used at each layer, except the values at each output are multiplied by k because k rotational variations of kernels are required. For example, the number of parameters in the first layer of group equivariant convolution with a $m \times n$ kernel size and l such kernels for input with l channels can be calculated as follows: $m \times n \times l \times k$. Weight sharing in group equivariant convolutional networks results in a reduction of the total number of parameters compared to

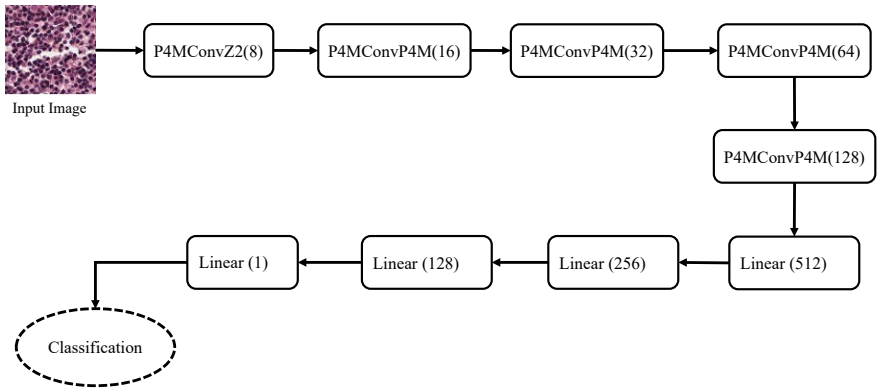


Figure 7: Overview of model architecture for tumor identification incorporating output channel details.

standard convolutional networks. This parameter reduction leads to improved efficiency in learning rotational variations, as the network can leverage shared weights across different orientations. By reducing the number of parameters, group equivariant convolutional networks can improve the accuracy of the network while reducing computational and memory costs.

3.1.1. Loss Function

Binary cross entropy[15] measures the similarity between predicted probabilities and actual binary outputs, which can only take values of 0 or 1. This metric penalizes the probabilities based on the deviation from the expected values, indicating how much they differ from the true binary labels. Specifically, it measures the distance between the predicted probabilities and actual binary outputs, thereby quantifying the performance of a binary classification model.

$$\mathcal{L} = C \cdot \left[-\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \right] \tag{1}$$

In this equation, C is the regularization parameter $(y_i, \hat{y}_i)$ are ground truth and predicted label of cancer respectively.

3.1.2. Evalution Matrix

The confusion matrix is a common evaluation metric used in classification models. It is a table that summarizes the performance of a classification model by counting the number of true positive, true negative, false positive, and false negative predictions made on the test set. Specifically, each element in the matrix represents the number of test samples that were predicted as belonging to each class, and whether that prediction was correct or incorrect. The confusion matrix is an essential tool for understanding the strengths and weaknesses of a classification model and can be used to compute other performance metrics such as sensitivity, specificity, and AUC scores.

Sensitivity : it refers to the true positive rate or the proportion of actual positives that are correctly identified as positives. Sensitivity can be represented mathematically as:

$$\text{Sensitivity} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Specificity: it refers to the true negative rate or the proportion of actual negatives that are correctly identified as negatives. Specificity can be represented mathematically as:

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}$$

AUC Score: it is a commonly used evaluation metric for binary classification models. It represents the area under the receiver operating characteristic (ROC) curve, which plots the true positive rate against the false positive rate at various classification thresholds.

The AUC score can be mathematically represented as:

$$\text{AUC} = \int_0^1 \text{TPR}(f)\, d\text{FPR}(f)$$

where TPR is the true positive rate and FPR is the false positive rate, and f

is the classification threshold. The integral represents the area under the ROC curve.

3.2. Model for Nucleus Instance Segmentation

The model's architecture features an encoder and decoder unit structure, where the encoder unit's output is integrated with two decoder units following a configuration akin to HoverNet[16]. Notably, the internal layers in this model comprise G-CNN layers, specifically P4mConvZ2 and P4mConvP4m. Two distinct branches, each preceded by encoders, are employed. In the first branch, the prediction focuses on nucleus pixel values, while the second branch anticipates the horizontal and vertical distances of the nucleus pixel from the center of mass (nucleus center). Following the post-processing of horizontal and vertical gradient maps, nucleus segmentation takes place alongside nucleus pixel predictions.

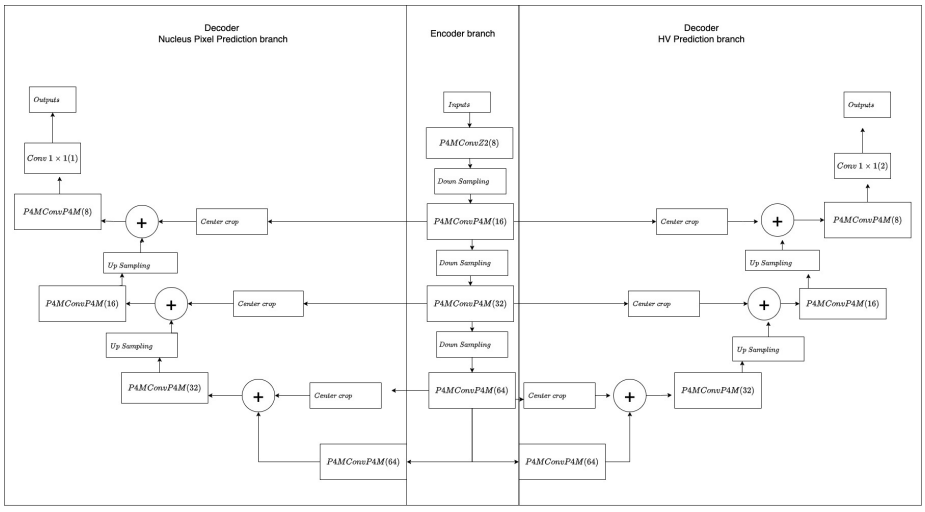


Figure 8: A comprehensive view of the model architecture, detailing inputs and outputs along with output channel information at each stage of both the encoder and decoder.

The architecture contains the outline of layers. Two branches used in the decoder part have the same up-sampling operations and P4MConvP4M layers.

this concept is similar to the densely connected units present in this paper [17]. In this architecture skip connection from encoder features is used at the decoder stage of each branch[18]. This helps to incorporate features from the encoders. A shared encoder helps to predict both nucleus pixels as well as the horizontal and vertical distance which will help in reducing overall training time.

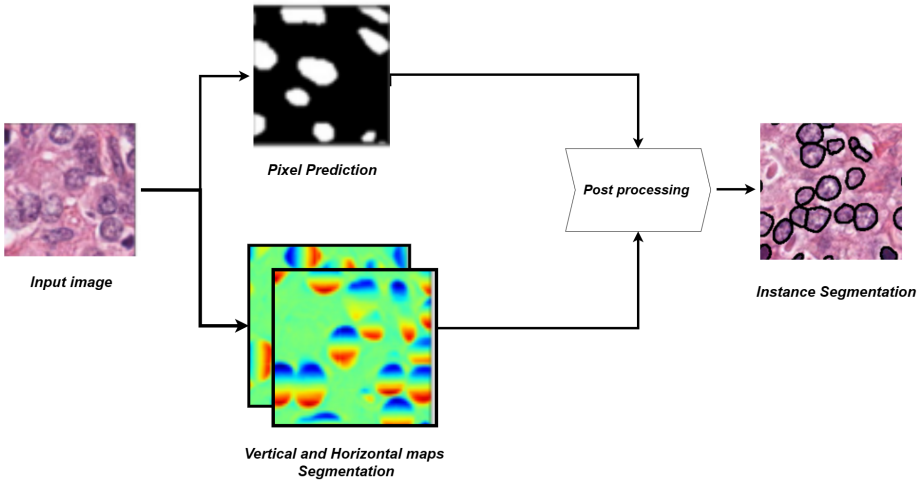


Figure 9: Post Processing step involves the calculation of vertical and horizontal gradient maps with energy landscapes and instance markers to produce the instance segmentation.

3.2.1. Loss Function

The Loss function is defined combination of regression loss:

$$\mathcal{L} = \underbrace{\lambda_a \mathcal{L}_a + \lambda_b \mathcal{L}_b}_{\text{Nucleus Pixel Prediction Branch}} + \underbrace{\lambda_c \mathcal{L}_c + \lambda_d \mathcal{L}_d}_{\text{H/V Branch}} \tag{2}$$

In this equation $\mathcal{L}_a$ and $\mathcal{L}_b$ will be determined by the dice loss of the nucleus prediction branch where λ_a *and* λ_b are the respective scalar Coefficient of loss for this branch. the $\mathcal{L}_c$ and $\mathcal{L}_d$ will be determined by the regression loss of HV(horizontal/vertical) distance prediction branch where λ_c *and* λ_d are the respective scalar Coefficient of loss.

$$\mathcal{L}_c = \frac{1}{n}\sum_{i=1}^{n}\left(p_i\left(I;\boldsymbol{w}\right) - \Gamma_i(I)\right)^2$$

$$\mathcal{L}_d = \frac{1}{m}\sum_{i\in M}\left(\nabla_x\left(p_{i,x}\left(I;\boldsymbol{w}\right)\right) - \nabla_x\left(\Gamma_{i,x}(I)\right)\right)^2 + \frac{1}{m}\sum_{i\in M}\left(\nabla_y\left(p_{i,y}\left(I;\boldsymbol{w}\right)\right) - \nabla_y\left(\Gamma_{i,y}(I)\right)\right)^2$$

In regression loss,∇_x and ∇_y denote the value of gradient in horizontal and vertical directions respectively, n and m denote a total number of nuclear pixels within the image and M denotes the set containing all nuclear pixels. I denotes the input image, i denotes the pixel values, w weights where $p_i(I;\boldsymbol{w})$ as the regression output of HV prediction branch, $\Gamma_i(I)$ as the ground truth values. Dice loss for the nucleus prediction branch is calculated as :

$$\text{Dice} = 1 - \frac{2\times\sum_{i=1}^{N}\left(B_i(I)\times A_i(I)\right)+\epsilon}{\sum_{i=1}^{N}B_i(I)+\sum_{i=1}^{N}A_i(I)+\epsilon} \tag{3}$$

In Dice loss equation A is the ground truth, B is the prediction value, N denotes the set containing all nuclear pixels, and ϵ is the smoothness constant which is set to 1e−3.

3.2.2. Post Processing

Variations in pixel intensities across different instances on both horizontal and vertical maps are substantial. Calculating the gradient proves valuable in delineating nuclei boundaries, as it highlights elevated values between adjacent nuclei when significant differences in pixel values exist. This methodology is particularly effective for identifying instances where pronounced disparages in pixel intensities occur.

$$S_m = max(H_x(p_x), H_y(p_y)) \tag{4}$$

In this equation, p_x and p_y are horizontal and vertical predictions at the output of the decoder branch where H_x and H_y are the gradients of the maps. Therefore, S_m represents areas where the pixel intensity difference is high within HV maps. The Instance Marker is computed for high values of S_m, and the

threshold is established for these values, taking into account the prediction.

$$Instant\ Marker\ M = \sigma(\tau(q,h) - \tau(S_m,k)) \tag{5}$$

In this equation, σ is a function that takes all positive integer values and sets all the negative values to zero. $\tau(x,y)$ is a threshold function that works on the values x and sets the value of y as 1 or 0 otherwise. The values of h and k are chosen in such a way that it will always predict the optimum nucleus segmentation results from the nucleus prediction branch and q is the probability map output of this branch. Energy Landscape calculation E,

$$E = [1 - \tau(S_m,k)] * \tau(q,h) \tag{6}$$

such that instance marker is used as the marker during marker-controlled watershed for instance segmentation of the nucleus where the split $\tau(q,h)$ is decided by M, given the E value.

3.2.3. Evaluation metrics

To measure the performance of Instance segmentation various parameters are available which include Ensemble DICE[19], Aggregate Jaccard Index AJI[12], Panoptic Quality PQ [20] here each of the matrices measures different aspects of segmentation.

DICE2 value is the dice value per nucleus, where dice and AJI are calculated as below equations respectively with A being the ground truth value and B being the prediction value.

$$DICE = 2 \times (A \cap B)/(|A| + |B|) \tag{7}$$

$$AJI = 2 \times (A \cap B)/(A \cup B) \tag{8}$$

To enhance interpretability, the panoptic quality metric is partitioned into two components: Detection Quality (DQ) and Segmentation Quality (SQ). It is important to note that SQ is only applicable within authentic positive segments and should be evaluated alongside DQ.

$$\mathcal{PQ} = \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\text{Detection Quality(DQ)}} \times \underbrace{\frac{\sum_{(A,B)\in TP} \text{IoU}(A,B)}{|TP|}}_{\text{Segmentation Quality(SQ)}} \tag{9}$$

All the pair value are mathematically unique proved by [20] if $IoU(A, B) > 0.5$. here Intersection Over Union (IoU) denotes the degree of overlap between the cluster of nuclei. Detection quality DQ is the F_1 score that is used for instance detection and segmentation quality SQ is used for how accurately an instance is being detected. Please refer to Appendix A for a full analysis of the loss function and post-processing method.

4. Results & Discussion

4.1. Identification of cancer

4.1.1. PCam Dataset

Training of model is done using a subset of the PCam dataset in which duplicate images had been removed. The PCam dataset is a 10x down-sampling of the Camelyon16 Challenge Dataset which contains 400 hematoxylin and eosin (H&E) stained whole slide images of sentinel lymph node sections, collected in the Radboud University Medical Center and University Medical Center Utrecht, and digitized at 2 different centers using a 40x objective[9].

In total, the project dataset consisted of 262,176 color images (96px x 96px) with a corresponding binary label indicating whether or not the image contained metastatic tissue. The class distribution is about a 50:50 split between negative and positive examples. the dataset is split into 80% training examples (209728), 10% validation examples (26,224), and 10% test examples (26,224).sped up in the learning process for the model by normalizing all input images with a mean and standard deviation of 0.5 for each channel.

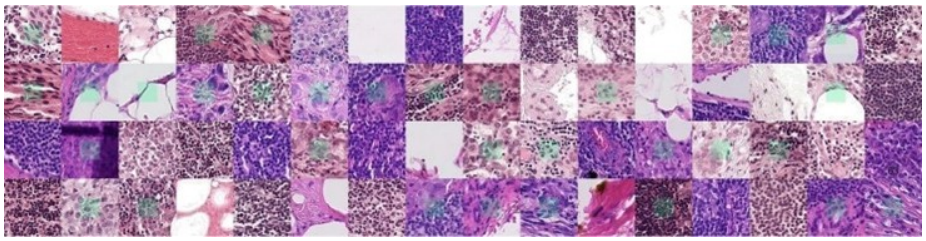


Figure 10: Histopathological image of tissue

4.1.2. Kumar Dataset

The Kumar dataset is comprised of 30 image tiles, each measuring 1,000 by 1,000 pixels and captured at a magnification of 40x[12]. The dataset includes samples from seven different organs, namely six breasts, six livers, six kidneys, six prostates, two bladders, two colons, and two stomachs. Additionally, each image tile in the dataset contains a complete marking of the boundary delineating the nucleus.

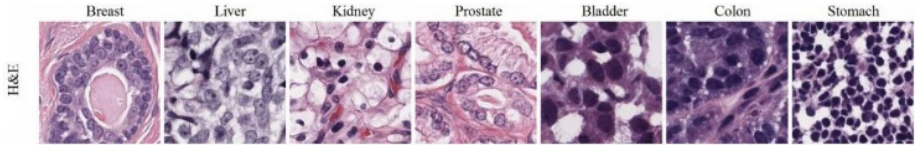


Figure 11: Histopathological images of Kumar Dataset

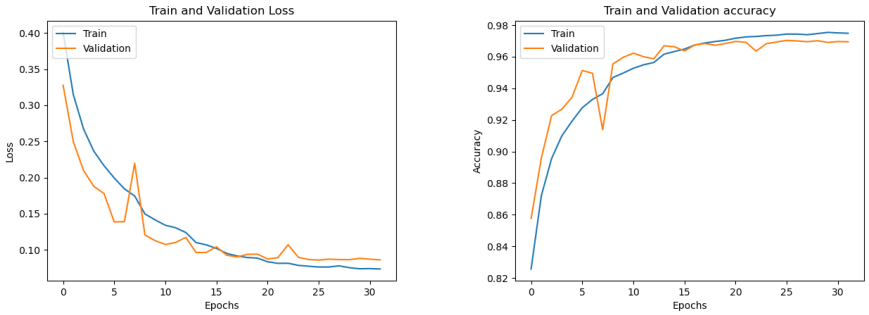
A total of ten test images are used to test our model based on the Group Equivariant Network. The instance nucleus segmentation detection on this test data.

4.1.3. Evaluation for Classification Tumor

The model achieves an accuracy of 98.4% on the Pcam dataset respectively. This is comparable to the current state-of-the-art model's performance. The result of the model prediction is presented below:

The optimum parameter was taken with initial learning rate $= 0.5 \times 10^{-}5$, Batch size $= 16$

The model was trained for 30 epochs with an early stopping technique[21] employed to identify the optimal loss value for the validation dataset. Early stopping is a regularization technique that halts training when the performance on a validation set starts to degrade, preventing overfitting and ensuring that the model generalizes well to unseen data. By using this technique, the model is trained efficiently and the risk of overfitting is reduced, resulting in a more accurate and robust model.



(a) Loss Vs Epoch

(b) Accuracy Vs Epoch

Figure 12: Training and Testing

The following is a confusion matrix representing the performance of the model on the test data of PCAM dataset :

N=26224	Actual Positive	Actual Negative
Predicted Positive	**TP = 12931**	FN = 225
Predicted Negative	FP = 205	**TN = 12853**

Table 1: Confusion Matrix

CNN based Model: In order to evaluate the performance of group equivariant layers, the architecture of the model was kept the same but all group equivariant layers were replaced with standard convolution layers. The model was evaluated using the same set of evaluation metrics, including accuracy, Sensitivity, specificity, AUC score, and confusion matrix. By comparing the performance of the model with group equivariant layers to the model with standard convolution layers, we can assess the impact of group equivariant layers on the model's performance and determine whether they are a necessary component for achieving optimal results.

The table below shows a comparison of the results obtained with our Group Equivariant Convolutional Neural Network (G-CNN) model and a Standard

Convolutional Neural Network (CNN) model:

Model	Accuracy	Sensitivity	AUC Score
CNN based model	0.962	0.951	0.945
G-CNN based model	0.984	0.982	0.988

Table 2: Comparison of G-CNN and CNN performance

The G-CNN model outperforms the standard CNN model in terms of accuracy, precision, recall, and AUC score. These results demonstrate that the use of group equivariant layers can lead to improved performance in convolutional neural networks, especially in tasks that involve rotational invariance.

Model	**Parameter**	**AUC**
CNN based model	564K	0.945
DSF-CNN[9]	**2.2M**	0.975
G-CNN based model(**Ours**)	2.3M	**0.988**

Table 3: Tumour classification results on the PCAM dataset with a similar parameter budget.

4.1.4. Evaluation for Nucleus Instance Segmentation

Segmented Images: The output images generated by our G-CNN model for nucleus segmentation in histopathological images demonstrate a clear delineation of nucleus boundaries. The segmentation accurately identifies nuclei across different types of tissue, including breast, liver, kidney, prostate, bladder, colon, and stomach tissue. The predicted output shows high consistency with ground truth annotations and achieves good performance on evaluation metrics such as DICE2, PQ, and AJI. These results demonstrate the effectiveness of our G-CNN model for accurate and robust segmentation of nuclei in histopathological images.

Comparitive Results: To evaluate the impact of group equivariant layers on the performance of a model for a segmentation problem, we replaced all the group equivariant layers in the model with standard convolutional layers while

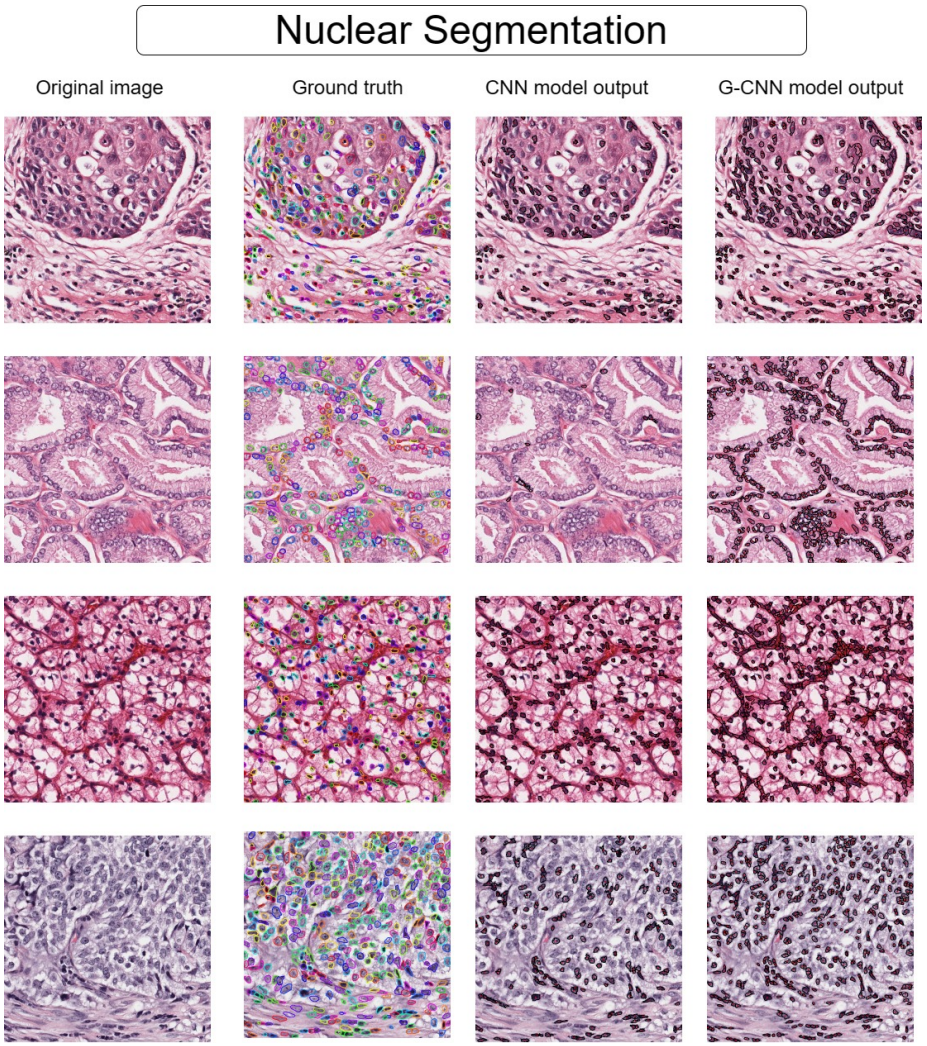


Figure 13: Nucleus Instance Segmentation with CNN and G-CNN Based Model

keeping the rest of the architecture the same. The resulting model was evaluated using the same set of evaluation metrics, namely DICE2, PQ, and AJI. By comparing the performance of the original model with group equivariant layers to the modified model with standard convolutional layers, we assess the contribution of group equivariant layers to the model's performance with achieved optimal results. The results are presented in the following table for the Kumar

dataset for Instance nucleus Segmentation:

Model	Dice	AJI	DQ	SQ	PQ
CNN based model	0.12	0.07	0.05	0.47	0.03
G-CNN based model	0.64	0.35	0.41	0.61	0.25

Table 4: Comparison of G-CNN and CNN performance

The results demonstrate that both models exhibit strong generalization capability, with high performance on the Kumar dataset. However, Model G-CNN shows the highest performance across all evaluation metrics, suggesting that the inclusion of rotational invariant features in this model may contribute to its superior performance.

5. Conclusion

The application of the concept of group equivariant convolution extends to analogous challenges in medical image analysis, particularly in tasks such as nucleus segmentation and classification. For instance, in the Kumar dataset, where the objective is to identify nucleus segmentation within images, the limited number of example points has prompted the use of densely connected networks and pre-trained models for segmentation, as observed in hovernet[16].

An alternative approach to address this challenge is utilizing group equivariant convolution layers within a ResNet-like architecture. This design possesses a notable capacity for weight-sharing, negating the necessity for a pre-trained model. This strategy offers a potential solution for nucleus segmentation in scenarios where the dataset size is constrained, presenting a more efficient and adaptable approach.

Author Contributions

Prof. Jayanta Mukhopadhyay contributed to the conceptualization, editing, and reviewing of the manuscript. Dr. Sudipta Ghosh, Ms. Ankita Chatterjee,

and Mr. Rahul Mehta were responsible for writing the manuscript, formal analysis, data curation, and methodology.

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Competing Interest

The authors declare no competing interest.

References

[1] M. W. Taco S. Cohen, Group equivarient convolution, Machine learning 10 (2016) 5–32.

[2] D. Albashish, R. Al-Sayyed, A. Abdullah, M. H. Ryalat, N. Ahmad Almansour, Deep cnn model based on vgg16 for breast cancer classification, in: 2021 International Conference on Information Technology (ICIT), 2021, pp. 805–810. doi:10.1109/ICIT52682.2021.9491631.

[3] K. Simonyan, A. Zisserman, Very deep convolutional networks for large-scale image recognition (2014). doi:10.48550/ARXIV.1409.1556.
URL https://arxiv.org/abs/1409.1556

[4] S. M. A. Alal Barou, Kun Zhan, Classification of breast cancer based on histology images using convolutional neural networks, IEEE ACCESS 6.

[5] M. Weiler, F. A. Hamprecht, M. Storath, Learning steerable filters for rotation equivariant cnns, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 849–858. doi:10.1109/CVPR.2018.00095.

[6] M. W. Lafarge, E. J. Bekkers, J. P. Pluim, R. Duits, M. Veta, Roto-translation equivariant convolutional networks: Application to histopathology image analysis, Medical Image Analysis 68 (2021) 101849. doi:https://doi.org/10.1016/j.media.2020.101849.
URL https://www.sciencedirect.com/science/article/pii/S1361841520302139

[7] M. W. Taco S. Cohen, Group equivariant convolutional networks (2016). doi:10.48550/arXiv.1602.07576.
URL https://arxiv.org/abs/1602.07576v3

[8] T. Cohen, Equivariant convolutional networks, in: Equivariant convolutional networks, 2021, pp. 0–245.

[9] S. Graham, D. Epstein, N. Rajpoot, Dense steerable filter cnns for exploiting rotational symmetry in histology images, IEEE Transactions on Medical Imaging 39 (12) (2020) 4124–4136. doi:10.1109/TMI.2020.3013246.

[10] B. S. Veeling, J. Linmans, J. Winkens, T. Cohen, M. Welling, Rotation equivariant cnns for digital pathology, in: Medical Image Computing and Computer Assisted Intervention–MICCAI 2018: 21st International Conference, Granada, Spain, September 16-20, 2018, Proceedings, Part II 11, Springer, 2018, pp. 210–218.

[11] B. E. Bejnordi, M. Veta, P. J. Van Diest, B. Van Ginneken, N. Karssemeijer, G. Litjens, J. A. Van Der Laak, M. Hermsen, Q. F. Manson, M. Balkenhol, et al., Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer, Jama 318 (22) (2017) 2199–2210.

[12] N. Kumar, R. Verma, S. Sharma, S. Bhargava, A. Vahadane, A. Sethi, A dataset and a technique for generalized nuclear segmentation for computational pathology, IEEE Transactions on Medical Imaging 36 (7) (2017) 1550–1560. doi:10.1109/TMI.2017.2677499.

[13] K. Tomczak, P. Czerwińska, M. Wiznerowicz, Review the cancer genome atlas (tcga): an immeasurable source of knowledge, Contemporary Oncology/Współczesna Onkologia 2015 (1) (2015) 68–77.

[14] N. Dave, V. Vinze, J. Dhami, N. Katre, S. Correia, Study of group equivariant convolutional networks for image classification, in: 2021 International Conference on Advances in Computing, Communication, and Control (ICAC3), 2021, pp. 1–5. doi:10.1109/ICAC353642.2021.9697298.

[15] C. M. Bishop, Pattern recognition and machine learning.

[16] S. Graham, Q. D. Vu, S. E. A. Raza, A. Azam, Y. W. Tsang, J. T. Kwak, N. Rajpoot, Hover-net: Simultaneous segmentation and classification of nuclei in multi-tissue histology images, Medical Image Analysis 58 (2019)

101563. doi:https://doi.org/10.1016/j.media.2019.101563.
URL https://www.sciencedirect.com/science/article/pii/S1361841519301045

[17] G. Huang, Z. Liu, L. van der Maaten, K. Q. Weinberger, Densely connected convolutional networks (2018). arXiv:1608.06993.

[18] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: N. Navab, J. Hornegger, W. M. Wells, A. F. Frangi (Eds.), Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015, Springer International Publishing, Cham, 2015, pp. 234–241.

[19] Q. D. Vu, S. Graham, M. N. N. To, M. Shaban, T. Qaiser, N. A. Koohbanani, S. A. Khurram, T. M. Kurç, K. Farahani, T. Zhao, R. R. Gupta, J. T. Kwak, N. M. Rajpoot, J. H. Saltz, Methods for segmentation and classification of digital microscopy tissue images, CoRR abs/1810.13230. arXiv:1810.13230.
URL http://arxiv.org/abs/1810.13230

[20] A. Kirillov, K. He, R. Girshick, C. Rother, P. Dollar, Panoptic segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019.

[21] M. Mahsereci, L. Balles, C. Lassner, P. Hennig, Early stopping without a validation set, CoRR abs/1703.09580. arXiv:1703.09580.
URL http://arxiv.org/abs/1703.09580